



Correlation and Regression · Worksheet 1

Statistics Chapter 4: Correlation and Regression. Answer all questions. Show your working clearly. Total: 86 marks. Time: 110 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are four sections, one for each topic in Chapter 4. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- Use $S_{xx} = \sum x^2 - \frac{(\sum x)^2}{n}$, $S_{xy} = \sum xy - \frac{\sum x \sum y}{n}$, $r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}$, $b = \frac{S_{xy}}{S_{xx}}$, $a = \bar{y} - b\bar{x}$.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

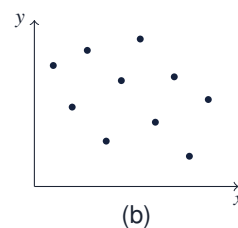
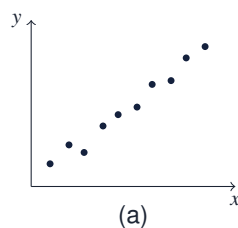
Section	Marks available	Marks scored
A Scatter diagrams and correlation	17	
B The product moment correlation coefficient	22	
C Linear regression	21	
D Non-linear models and logarithms	21	
Roar-level challenge	5	
Total	86	

Section A Scatter diagrams and correlation

17 marks

1. Describe the type of correlation shown in each scatter diagram.

[2]





2. A teacher records the number of hours of revision, h , and the test score, s , of each student in her class. [2]

State, with a reason, which is the explanatory variable and which is the response variable.

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3. Data collected over a year show strong positive correlation between the daily number of ice creams sold at a beach and the daily number of people rescued by lifeguards. [3]

A newspaper claims that eating ice cream causes people to get into difficulty in the sea. Comment on this claim.

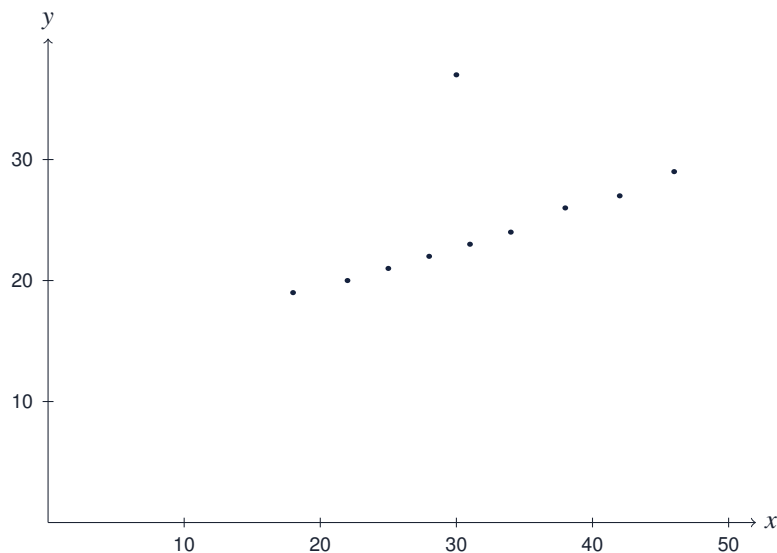
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4. The scatter diagram shows the ages, x years, and the 5 km run times, y minutes, of 10 club runners. [3]



- (a) Identify the outlier on the diagram and suggest a reason for it.
- (b) Ignoring the outlier, describe the correlation and interpret it in context.

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5. A student uses the large data set to draw a scatter diagram of daily mean temperature, t °C, against daily total sunshine, s hours, for Heathrow in June 2015. The data show weak positive correlation. The values of s range from 0.4 to 14.2.

(a) Interpret the correlation in context.

[2]

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(b) State, with a reason, which variable is the explanatory variable.

[1]

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(c) The student wants to use the data to estimate the daily mean temperature on a day in June with 18 hours of sunshine. Give two reasons why this may not be reliable.

[2]

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(d) The student concludes that “more sunshine causes higher temperatures in every month”. Give two criticisms of this conclusion.

[2]

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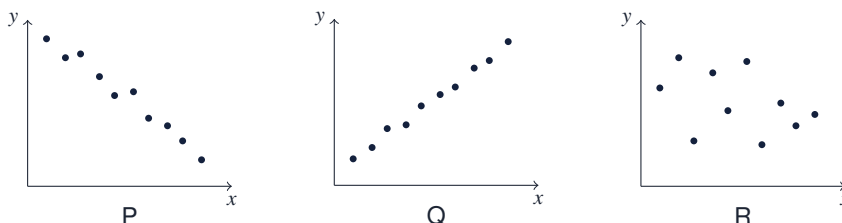
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(Total for Question 5 is 7 marks)

**Section B The product moment correlation coefficient**

22 marks

6. Match each of the values $r = 0.9$, $r = -0.2$ and $r = -0.85$ to one of the scatter diagrams P, Q and R. [3]



7. For a set of 6 pairs of data, $\sum x = 30$, $\sum y = 48$, $\sum x^2 = 190$, $\sum y^2 = 436$, $\sum xy = 268$. [4]
Find S_{xx} , S_{yy} and S_{xy} , and hence calculate the product moment correlation coefficient.

8. The product moment correlation coefficient between x and y is 0.73. [3]

Write down the product moment correlation coefficient between

(a) $u = x - 10$ and $v = \frac{y}{5}$,

(b) $u = 2x + 3$ and $w = 100 - y$.

Explain your answers.



9. Calculate the product moment correlation coefficient for the data below.

[4]

x	1	2	3	4	5
y	3	5	4	8	10

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10. The number of hours, x , spent on an online maths course and the final test score, y , of 8 students are summarised by

$$\sum x = 48, \quad \sum y = 320, \quad \sum x^2 = 340, \quad \sum y^2 = 13\,400, \quad \sum xy = 2040.$$

- (a) Find S_{xx} , S_{yy} and S_{xy} .

[3]

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- (b) Calculate the product moment correlation coefficient between x and y .

[2]

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- (c) Interpret your answer in context.

[1]

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- (d) The test scores are converted to a new scale by multiplying by 2.5. State, with a reason, the product moment correlation coefficient between x and the new scores.

[2]

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(Total for Question 10 is 8 marks)



Section C Linear regression

21 marks

11. The regression line of y (the cost of a job, in £10s) on x (the number of hours worked) is $y = 3.2 + 1.5x$. [3]

- (a) Interpret the value 1.5 in context.
(b) Interpret the value 3.2 in context.
(c) Estimate y when $x = 4$.

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12. For a set of data, $S_{xx} = 40$, $S_{xy} = 28$, $\bar{x} = 5$ and $\bar{y} = 8$. [3]

Find the equation of the regression line of y on x in the form $y = a + bx$.

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13. Using the data in Question 9, find the equation of the regression line of y on x . [3]

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14. The regression line of y on x for a set of data with x values between 10 and 40 is $y = 12 + 0.8x$. [3]

Estimate y when (a) $x = 25$, (b) $x = 60$. Comment on the reliability of each estimate.

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15. The age, x years, and the value, y thousand pounds, of 10 used cars of the same model are summarised by

$$\sum x = 50, \quad \sum y = 85, \quad \sum x^2 = 310, \quad \sum xy = 360.$$

- (a) Find the equation of the regression line of y on x . [4]

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- (b) Interpret the gradient of your line in context. [1]

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- (c) Estimate the value of a car of this model that is 4 years old. [2]

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- (d) Use your line to estimate the value of a 15-year-old car, and comment on your answer. [2]

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(Total for Question 15 is 9 marks)



Section D Non-linear models and logarithms

21 marks

16. State the graph you would plot to obtain a straight line if (a) $y = ax^n$, (b) $y = kb^x$. [2]

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17. A graph of $\log_{10} y$ against $\log_{10} x$ is a straight line with gradient 1.5 and intercept 0.48 on the vertical axis. [3]

Find y in the form $y = ax^n$, giving a to 3 significant figures.

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18. The relationship between x and y is modelled by $\log_{10} y = 0.2 + 0.3x$. [3]

Write this in the form $y = kb^x$, giving k and b to 3 significant figures.

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19. A graph of $\log_{10} y$ against x is a straight line passing through the points $(1, 0.7)$ and $(5, 1.9)$. [4]

(a) Show that $y = kb^x$ and find k and b to 3 significant figures.

(b) Estimate y when $x = 3$.

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20. The number of bacteria, N , in a culture t hours after the start of an experiment is recorded for $0 \leq t \leq 6$. A graph of $\log_{10} N$ against t gives a straight line with equation $\log_{10} N = 2.1 + 0.15t$.

(a) Show that $N = kb^t$, stating the values of k and b to 4 significant figures. [4]

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(b) Interpret the value of b in context. [2]

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(c) Estimate the number of bacteria after 10 hours, and comment on the reliability of your estimate. [3]

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(Total for Question 20 is 9 marks)



ROAR-LEVEL CHALLENGE

5 marks

21. Data are coded using $u = x - 50$ and $v = \frac{y - 200}{10}$. The regression line of v on u is $v = 1.2 + 0.4u$, [5]
and the product moment correlation coefficient between u and v is 0.82.
Find the equation of the regression line of y on x , and write down the product moment correlation
coefficient between x and y .

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END OF WORKSHEET

Total: 86 marks